

CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ SOLUTIONS

Chapter 2: Polynomials

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Zeroes and coefficients of a quadratic polynomial

1. If the sum of zeroes of the polynomial $p(x) = 2x^2 - k\sqrt{2}x + 1$ is $\sqrt{2}$, then the value of k is: (a) $\sqrt{2}$ (b) 2 (c) $2\sqrt{2}$ (d) $\frac{1}{2}$ **CBSE 2024, 1 mark**

SOLUTION

$$\text{Sum of zeroes} = -\frac{b}{a} = \frac{k\sqrt{2}}{2}$$

$$\frac{k\sqrt{2}}{2} = \sqrt{2} \text{ gives } k = 2$$

Answer: (b)

2. The zeroes of a polynomial $x^2 + px + q$ are twice the zeroes of the polynomial $4x^2 - 5x - 6$. The value of p is: (a) $-\frac{5}{2}$ (b) $\frac{5}{2}$ (c) -5 (d) 10 **CBSE 2024, 1 mark**

SOLUTION

Let the zeroes of $4x^2 - 5x - 6$ be α and β . Then $\alpha + \beta = \frac{5}{4}$.

The zeroes of $x^2 + px + q$ are 2α and 2β , with sum $2(\alpha + \beta) = \frac{5}{2}$.

$$\text{Sum of zeroes} = -p, \text{ so } p = -\frac{5}{2}$$

Answer: (a)

2 Zeroes from the graph

3. Assertion (A): If the graph of a polynomial touches the x -axis at only one point, then the polynomial cannot be a quadratic polynomial. Reason (R): A polynomial of degree n ($n > 1$) can have at most n zeroes. Choose: (a) Both A and R are true and R is the correct explanation of A. (b) Both A and R are true but R is not the correct explanation of A. (c) A is true but R is false. (d) A is false but R is true. **CBSE 2024, 1 mark**

SOLUTION

A is false: the graph of $(x - 1)^2 = x^2 - 2x + 1$ touches the x -axis only at $x = 1$, and it is a quadratic polynomial.

R is true: a polynomial of degree n has at most n zeroes.

Answer: (d)

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